

Question 1:

A fair die is rolled. If an odd number is obtained, then that is the recorded score. However, if an even number is obtained, then a fair coin is tossed. If a head is obtained, then the recorded score is the number on the die, but if a tail is obtained, then the recorded score is half of the number on the die.

- (a) Give the possible values of the recorded score.
- (b) What is the probability that a score of five is recorded?
- (c) What is the probability that a score of three is recorded?

x_i	1	2	3	4	5	6
head	1	2	3	4	5	6
tail	1	1	3	2	5	3

(a) $1, 2, 3, 4, 5, 6$

(b) $P(5) = \frac{1}{6}$

(c) $P(3) = \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{2} = \frac{3}{12} = \frac{1}{4}$

Question 2:

The marketing division of a company profiles its potential customers and grades them as either likely or unlikely purchasers. Overall, 16% of the potential customers are graded as likely purchasers. In reality, 81% of the potential customers graded as likely purchasers actually make a purchase, while only 9% of the potential customers graded as unlikely purchasers actually make a purchase. If somebody made a purchase, what is the probability that they had been graded as a likely purchaser?

$$P(e) = 0.16$$

$$P(p|e) = 0.81$$

$$P(e') = 0.84$$

$$P(p|e') = 0.09$$

$$P(e|p) = \frac{P(e) \cdot P(p|e)}{P(e) \cdot P(p|e) + P(e') \cdot P(p|e')} =$$

$$= \frac{0.16 \cdot 0.81}{0.16 \cdot 0.81 + 0.84 \cdot 0.09} = \frac{0.1296}{0.2052} = 0.6316$$

Question 3:

A random variable X takes values between 0 and 4 with a cumulative distribution function

$$F(x) = \frac{x^2}{16}, \quad 0 \leq x \leq 4.$$

- (a) What is $P(X \leq 2)$?
- (b) What is $P(1 \leq X \leq 3)$?
- (c) Find the probability density function.
- (d) What is the expected value of this random variable?
- (e) What is the median of this random variable?
- (f) What is the variance of this random variable?
- (g) What is the standard deviation of this random variable?

$$(a) P(X \leq 2) = F(2) = \frac{4}{16} = \frac{1}{4}$$

$$(b) P(1 \leq X \leq 3) = F(3) - F(1) = \frac{9-1}{16} = \frac{1}{2}$$

$$(c) f(x) = \frac{dF}{dx} = \frac{x}{8}, \quad 0 \leq x \leq 4$$

$$(d) E(X) = \int_0^4 \frac{x^2}{8} dx = \frac{x^3}{24} \Big|_0^4 = \frac{4^3}{24} = \frac{8}{3}$$

$$(e) F(x) = 0.5 \Rightarrow \frac{x^2}{16} = 0.5 \Rightarrow x = 2\sqrt{2}$$

$$(f) E(X^2) = \int_0^4 \frac{x^3}{8} dx = \frac{x^4}{32} \Big|_0^4 = \frac{4^4}{32} = 8$$

$$\text{Var}(X) = 8 - \left(\frac{8}{3}\right)^2 = 8 - \frac{64}{9} = \frac{72-64}{9} = \frac{8}{9}$$

$$(g) \sigma = \frac{2\sqrt{2}}{3}$$

Question 4:

A fair coin is tossed two times, and the random variable X is the number of heads in the two tosses and the random variable Y is the number of heads in the second toss.

- What is the joint probability mass function of X and Y ?
- What are the marginal probability mass functions of X and Y ?
- Are the random variables X and Y independent?
- What are the expectations and variances of the random variables X and Y ?
- What is the covariance of X and Y ?
- If there is one head in the second toss, what is the conditional probability mass function of X ? What is the conditional expectation of X ?

$Y \backslash X$	0	1	2	P_{+j}
0	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	0	$\frac{1}{2}$
1	0	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	$\frac{1}{2}$
P_{i+}	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1

(a), (b) see the table

(c) no, they are not independent

$$P_{32} \neq P_3 \cdot P_2$$

$$\frac{1}{4} \neq \frac{1}{4} \cdot \frac{1}{2}$$

$$(d) E(X) = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1, E(X^2) = 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}; \text{Var}(X) = \frac{1}{2}$$

$$E(Y) = \frac{1}{2}; E(Y^2) = \frac{1}{2} \Rightarrow \text{Var}(Y) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$(e) \text{cov}(X, Y) = E(XY) - E(X)E(Y) = 1 \cdot 1 \cdot \frac{1}{4} + 1 \cdot 2 \cdot \frac{1}{4} = \frac{3}{4}$$

$$(f) P(X=0|Y=1) = \frac{P_{01}}{P_{+1}} = 0$$

$$P(X=1|Y=1) = \frac{P_{11}}{P_{+1}} = \frac{1/4}{1/2} = \frac{1}{2}$$

$$P(X=2|Y=1) = \frac{P_{21}}{P_{+1}} = \frac{1/4}{1/2} = \frac{1}{2}$$

$$E(X|Y=1) = 0 \cdot 0 + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2}$$

Question 5:

A biologist has a culture consisting of 6 cells. In a period of 1 hour, independent of the other cells, there is a probability of 0.4 that each of these cells splits into 2 cells.

- (a) Which distribution is applicable in this problem, justify.
- (b) What is the probability that after 1 hour the biologist has at least 10 cells?
- (c) What is the expected number of cells after 1 hour?

(a) In this case we have a binomial distribution $B(6, 0.4)$ because each of the cells has just two ways: splits or does not split.

$$(b) P(X \geq 4) = P(X=4) + P(X=5) + P(X=6) \\ = C_4^6 0.4^4 \cdot 0.6^2 + C_5^6 0.4^5 \cdot 0.6^1 + C_6^6 0.4^6 \cdot 0.6^0$$

$$(c) E(X) = np + 6 = 6 \cdot 0.4 + 6 = 8.4$$

Question 6:

Suppose that you are waiting for a friend to call you and that the time you wait in minutes has an exponential distribution

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

and $f(x) = 0$ for $x < 0$ with parameter $\lambda = 0.2$. The cumulative distribution function is

$$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

and $F(x) = 0$ for $x < 0$.

- (a) What is the expectation of your waiting time?
- (b) What is the probability that you will wait longer than 10 minutes?
- (c) What is the probability that you will wait less than 5 minutes?
- (d) Suppose that after 5 minutes you are still waiting for the call. What is the distribution of your additional waiting time? In this case, what is the probability that your total waiting time is longer than 10 minutes?

(a) $E(X) = \frac{1}{\lambda} = 5$

(b) $P(X > 10) = 1 - F(10) = 1 - (1 - e^{-0.2 \cdot 10}) = e^{-1}$

(c) $P(X < 5) = F(5) = 1 - e^{-0.2 \cdot 5} = 1 - e^{-1}$

(d) After 5 minutes we still have the same distribution $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$, $\lambda = 0.2$.

$P(X > 10) = 1 - F(10) = 1 - (1 - e^{-0.2 \cdot 10}) = e^{-1}$